The Impulse Protocol: Non-Destructive Coupling Discovery in Autonomous Agents

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Abstract

Multiple autonomous agents tending a shared substrate—physical, computational, or mathematical—unknowingly affect each other through coupling channels invisible to each agent individually. We present the impulse protocol, a method by which such agents discover their coupling structure through coordinated perturbation: one agent deliberately drives its parameters to extremes (push), then returns to neutral (pause), while all others hold fixed as passive sensors. An adaptive perturbation calibration phase finds a safe amplitude before each push block. A constant-false-alarm-rate (CFAR) step-change detector then evaluates whether each receiver's signal exhibits a reversible shift correlated with the sender's perturbation.

The protocol is coupling-shape-agnostic—it does not model the coupling function and detects linear, threshold, polynomial, and saturated coupling equally. We characterize the method empirically on a synthetic bench with planted ground truth, sweeping coupling strength (0.1–0.9), noise level ($\sigma = 0.01$–0.20), and coupling shape. The detection floor lies between coupling 0.1 and 0.2 at noise $\sigma = 0.02$, with measured step amplitudes within 5% of planted ground truth. Zero false positives were observed across all noise levels tested.

The contribution is the protocol that makes coupling *detectable*—not a novel detector. The detection backend is pluggable. The key insight is that autonomous agents, unaware of each other and communicating only through the substrate they tend, can be coordinated to take turns as active perturbers and passive observers through a shared lease—transforming an unresolvable observational confound into a structured experiment.

Keywords: active probing, coupling discovery, autonomous agents, impulse protocol, CFAR detection, multi-agent coordination, non-destructive perturbation, cybernetics

1 Introduction

Complex coupled systems—data centers, microgrids, process plants, distributed databases, microservice architectures, compiler pipelines—share a fundamental problem: their internal coupling structure is hidden. Which subsystems affect which others, through which channels, at what strength? The design documentation specifies intended interfaces, but operational reality includes emergent coupling paths created by the shared substrate that no blueprint captures and no expert enumerates [7].

Three existing approaches all fail to discover this structure in live systems. Analytical modeling requires knowing the system's structure a priori. Statistical observation—cross-correlation, Granger causality, transfer entropy—cannot separate causal coupling from co-movement when both subsystems are actively adjusting, because the dominant noise source is the agents' own exploration [9, 10]. Expert systems capture design intent, not operational reality, and go stale as the system drifts.

The confounding problem. Consider multiple autonomous agents, each independently tending a portion of a shared substrate. Each agent adjusts its own parameters toward its own objective. The agents are unaware of each other. Yet they communicate involuntarily through the substrate: when one agent changes its parameters, the system responds in ways that shift other agents' readings.

When all agents are actively tending, their objective readings move for two confounded reasons: their own parameter changes and the other agents' influence. No passive observer can separate these contributions. The confounding is structural, not statistical—no amount of observational data resolves it without intervention. This is why cross-correlation, Granger causality, and transfer entropy all fail on actively explored systems. They cannot distinguish "shifting together because coupled" from "shifting together because each is independently adjusting."

The key insight. The confound disappears if one agent stops exploring and holds still while another perturbs. With the receiver frozen, any change in its signal is attributable to the sender's perturbation—not to its own adjustments. The agent ceases to be an explorer and becomes a sensor. The sender ceases to explore and becomes a stimulus.

This is the active probing paradigm, borrowed from radar and seismology: inject a known signal into the medium and observe the response. The system describes itself through its response to perturbation, rather than through passive observation of its behavior. The coupling graph is not inferred from correlations—it is measured as a set of impulse responses.

Our approach. We embed this paradigm inside the agents' normal operation through a coordinated push/-pause/hold protocol. The protocol directs the agents to take turns: one agent (the sender) drives its parameters to extremes, then returns to neutral, while all others hold fixed as passive sensors. An adaptive calibration phase finds a safe perturbation amplitude before each push block, backing off exponentially if the system destabilizes. A constant-false-alarm-rate (CFAR) step-change detector evaluates whether each receiver's signal exhibits a reversible shift—rising during push, recovering during pause. The ABA block design eliminates drift confounds: a monotonic trend produces a rising edge but not a falling edge.

The protocol requires no information exchange beyond scheduling ("whose turn is it"), enforced through a shared lease with fencing tokens. Agents do not share objectives, results, or detected edges. Each discovers coupling purely by watching its own signal shift when another agent perturbs the substrate.

Coupling-shape agnosticism. Because the protocol measures step response—not frequency response, not correlation structure—it is agnostic to the coupling function's shape. Linear coupling produces a proportional step. Threshold coupling produces a discontinuous jump. Saturation coupling produces a compressed or inverted step. Polynomial coupling produces a quadratic step. The detector asks only: did the signal change and recover? The shape of the response determines the step amplitude but not whether coupling exists.

Contributions.

1. A coordination protocol—push, pause, hold, with adaptive perturbation calibration and turn-taking through a fencing-token lease—that enables autonomous agents to discover coupling structure through their shared substrate without exchanging information beyond scheduling.
2. Empirical characterization on a synthetic bench with planted ground truth across four coupling shapes (linear, threshold, saturation, polynomial), five noise levels, and six coupling strengths. Detection floor, noise boundary, and false positive rate are measured and reported honestly.
3. Step amplitude measurements within 5% of ground truth across the detectable range, with zero false positives across all noise levels tested.

The contribution is the protocol that makes coupling *detectable*—not a novel detector. CFAR is established technology from radar signal processing [4]. What is new is the choreography: autonomous agents, unaware of each other and communicating only through the substrate they tend, taking turns as active perturbers and passive observers. The agent that was exploring blindly becomes a measurement instrument. The probe *is* the experiment.

2 Related Work

The push/pause/hold protocol draws on active-probing paradigms from several fields where injecting a known signal into a medium and observing the response is the standard approach to discovering hidden structure. At its root, this is cybernetics—the study of control and communication in coupled systems [12]. Ashby's law of requisite variety [1] established that a controller must model its environment's complexity to act effectively; our protocol is a mechanism for discovering that complexity empirically when it cannot be specified analytically.

Radar and sonar. Constant-false-alarm-rate detection was developed for radar signal processing [4]. A transmitter emits a known waveform; a receiver listens for echoes. The CFAR threshold adapts to local noise conditions, maintaining a constant false-alarm probability regardless of background noise level. Active sonar applies the same principle underwater: a sound pulse propagates through the medium, and the return signal reveals structure (seafloor, submarines, fish schools) that passive listening cannot detect. Our protocol applies this paradigm to coupled systems: the push is the transmitted pulse, the receiver's objective is the return signal, and CFAR discriminates the coupling step from intrinsic noise.

Seismology. Controlled-source seismology detonates explosives or vibrates the ground at known locations, then measures ground motion at distant sensors. The subsurface structure—rock layers, fault lines, fluid reservoirs—is inferred from how the seismic waves propagate and reflect. The structure cannot be observed passively; it must be excited. Our method follows the same logic: coupling structure is inferred from how a perturbation propagates from one agent to another through the substrate.

Functional brain imaging (fMRI). In functional MRI, researchers study brain activity by presenting controlled stimuli and measuring the hemodynamic response—the slow, sustained blood-oxygen change that follows neural activation. The hemodynamic response function (HRF) is a low-pass filter that smears brief neural events into sustained signal changes. This led to the *block design* paradigm [5]: sustained stimulation blocks of 20–30 seconds alternating with rest periods, rather than single brief events. Our push/pause blocks are the same design: a sustained perturbation block that survives the substrate's thermal mass or response latency, followed by a pause block that captures recovery. The ABA structure (stimulus, rest, stimulus) is standard practice in experimental psychology and neuroscience [2].

Causal discovery. Pearl's do-calculus [8] provides the theoretical framework for causal inference from interventions. Convergent cross-mapping [10] infers coupling from time series but fails when systems are actively exploring. Neural relational inference [6] learns interaction graphs from trajectories but operates passively. Our method differs in that the interventions are performed by autonomous agents on themselves, coordinated through a shared lease, rather than by a central researcher.

Network tomography. Network tomography [11] infers internal network properties from end-to-end measurements using active probe packets. Active topology inference [3] selects measurement paths adaptively to reduce uncertainty about network structure. These approaches share our active-probing philosophy but assume a single centralized prober with an explicit identification objective. Our agents are autonomous—each with its own objective—and the probing is a cooperative side effect of coordination, not the agents' primary goal.

3 Method

3.1 System Model

We consider N autonomous agents (*tenders*), each tending a portion of a shared substrate. Agent i has a parameter vector $\mathbf{p}_i \in \mathbb{R}^{d_i}$ and an objective function f_i toward an objective. The substrate couples the agents: agent i's objective reading depends not only on its own parameters but also on other agents' parameters through hidden coupling channels:

$$f_i(\mathbf{p}_i, \mathbf{p}_{j\neq i}) = g_i(\mathbf{p}_i) + \sum_{j\neq i} c_{ji}(\mathbf{p}_j) \tag{1}$$

where g_i is the agent's own response and c_{ji} is the coupling contribution from agent j to agent i. The coupling function c_{ji} may be linear, nonlinear, or discontinuous. The coupling structure—which agents affect which, through which channels, at what strength—is unknown. Discovering it is the goal.

3.2 Agent Operation

Each agent runs a continuous trial loop. In normal operation, the agent samples a parameter vector via a metaheuristic search strategy (e.g., TPE, CMA-ES, NSGA-II), applies it to the substrate, reads back its objective, and records the result. This is the `OPTIMIZE` state—the agent's default behavior.

The coordination protocol interrupts this loop periodically to run detection rounds. Crucially, the perturbations during detection operate on the *same parameters* the agent controls during normal operation—the same knobs, within the same defined bounds. Safety is actively enforced through guardrails: per-trial checks on objective values and system state that reject configurations exceeding safe operating limits. The IMPULSE_CALIB phase probes at decreasing scale and backs off immediately on any guardrail violation. Only an amplitude that passes all guardrails is locked for the push block.

The protocol does not inject faults, disrupt services, or push the system outside its operating envelope. It adjusts known control parameters to the edge of their range, observes the response, and returns to neutral. The system continues operating throughout; detection is non-destructive. This distinguishes the approach from chaos engineering or fault injection, which deliberately break components to test resilience.

3.3 The Impulse Protocol

3.3.1 Sender State Machine

A detection round for the sender consists of eight states:

1. **`OPTIMIZE`**: Normal parameter exploration. After a minimum number of completed trials (≥ 15) and at least two active agents, the agent attempts to acquire the sender lease.

2. **`HOLD_CALIB`**: The sender establishes neutral hold parameters—values where its own objective is stable and the system is alive. When hold parameters are provided in configuration, this step is immediate. Otherwise, the sender searches for flat parameters by sampling its objective at candidate neutral values and accepting when the standard deviation falls below a threshold ($\sigma < 0.05$).

3. **`IMPULSE_CALIB`**: The sender probes at decreasing perturbation scale to find a safe amplitude. Starting at full scale (1.0), it drives its top-3 parameters (by range) toward their upper bounds. If guardrails are violated (e.g., the system destabilizes), it backs off: $1.0 \to 0.5 \to 0.25 \to 0.125$. The first scale that passes becomes the locked amplitude for the entire push block. This adaptive calibration ensures the perturbation is large enough to produce a detectable signal but not so large as to harm the live system.

4. **`PUSH`**: The sender perturbs at the locked scale for B_{push} consecutive trials. Each trial applies the extreme parameters, reads back, and records the objective. The coupling signal propagates through the substrate to any coupled receivers.

5. **PAUSE**: The sender returns to neutral hold parameters for B_{pause} consecutive trials. This is the ABA recovery—the coupling signal should revert. The receiver's signal during this window serves as the post-intervention baseline.

6. **DONE**: The sender releases the lease and clears coordination state.

7. **COOLDOWN**: The sender waits a short period (5 trials) before re-acquiring, giving other agents a chance to take their turn. This enforces fairness in turn-taking.

8. **HOLD** (receiver): While any sender holds the lease, all other agents enter HOLD—fixing their parameters at neutral values and acting as passive sensors. Each hold trial is tagged with the sender's current lease phase, so the detector can later segment receiver observations into push-window and pause-window samples.

3.3.2 *Lease and Turn-Taking*

Only one agent may be sender at any time. A shared lease (row in a coordination database) enforces this. The lease uses:

Fencing tokens A monotonically increasing token is written on each lease acquisition. Every lease mutation (heartbeat, budget decrement, release) is guarded by a WHERE `token = ` t check. If another agent steals the lease (via stale recovery), the old sender's token no longer matches and its writes are silently rejected—preventing split-brain.

Count-based budget The lease tracks `push_remaining` and `pause_remaining` counters rather than wall-clock expiry. The sender's turn lasts exactly as long as it has budget, avoiding premature expiry when trials run slowly (database retries, slow effectuation).

Heartbeat staleness A crashed sender's lease is recovered when its heartbeat exceeds 5× the worst-case trial duration. Budget alone is insufficient because a crashed sender's counters stay positive forever.

3.4 CFAR Step-Change Detection

After rounds complete, the detector evaluates each ordered sender–receiver pair. For each pair, the receiver's hold-trial objectives are segmented by the sender's push/pause timestamps into:

- *Baseline window*: receiver values during the sender's pause phase (post-intervention recovery)
- *Push window*: receiver values during the sender's push phase

A noise-adaptive threshold is computed from the baseline:

$$T = k \cdot \widehat{\sigma}_{\text{baseline}}, \quad k = N_{\text{ref}} \left(P_{\text{fa}}^{-1/N_{\text{ref}}} - 1 \right) \tag{2}$$

where $\widehat{\sigma}$ is the scaled median absolute deviation (MAD × 1.4826, a robust σ-estimator), N_{ref} is the number of baseline samples, and $P_{\text{fa}} = 0.05$.

A round is *detected* if both:

$$|\text{push\_median} - \text{baseline\_median}| \geq T \tag{3}$$

$$|\text{push\_median} - \text{pause\_median}| \geq T \tag{4}$$

Condition (3) detects the step; condition (4) enforces reversibility. A monotonic drift would satisfy (3) but fail (4) because the pause-phase values would not revert to baseline.

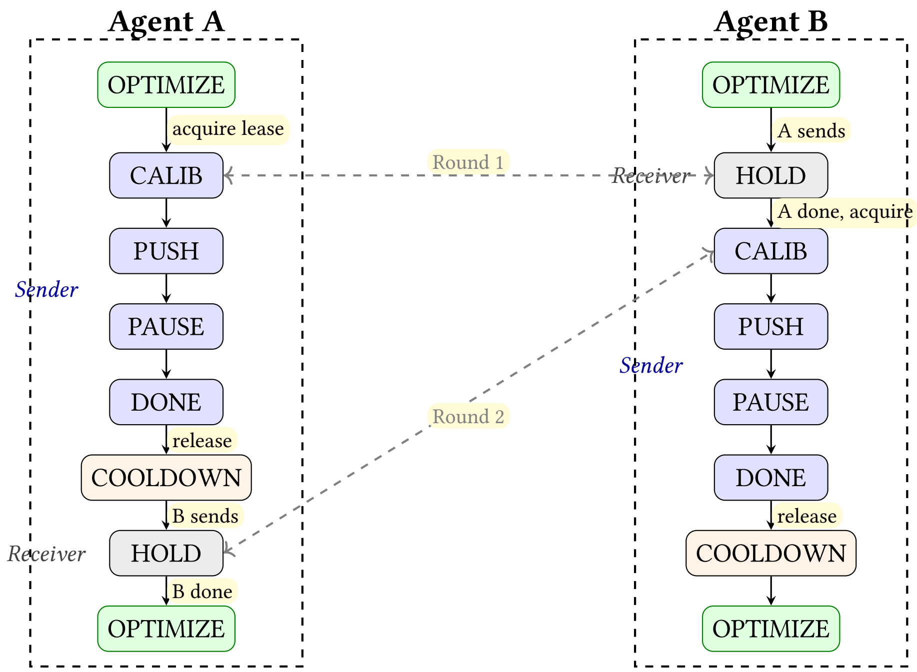

Figure 1: Two-agent turn-taking in the impulse protocol. Both agents start in OPTIMIZE. In Round 1, Agent A acquires the sender lease and enters the detection cycle: CALIB (hold neutral, find safe perturbation amplitude), PUSH (drive parameters to extremes), PAUSE (return to neutral), DONE (release lease). Agent B observes the active lease and enters HOLD as passive receiver throughout A's detection cycle. After cooldown, the roles reverse for Round 2. CALIB encompasses HOLD-CALIB (parameter flatness search) and IMPULSE-CALIB (adaptive perturbation with guardrail backoff). Dashed arrows show lease pairing: while one agent holds the sender lease, the partner is in HOLD.

3.5 Observation Parameters

All experiments use the following parameters unless otherwise stated:

Parameter	Value
Push block size (B_{push})	15 trials
Pause block size (B_{pause})	15 trials
Min. exploration before detection	15 trials
Cooldown (turn-taking fairness)	5 trials
Impulse calibration scales	1.0, 0.5, 0.25, 0.125
Hold parameters	midpoint (50, 50, 50)
Push target	top-3 params → upper bound
CFAR P_{fa}	0.05
Min. reference cells (baseline)	3
Min. test cells (push/pause)	3

Detection boundaries are parameterized by observation budget. Increasing block sizes or the number of rounds lowers the boundary, as the CFAR scale factor k decreases with more reference cells. The impulse calibration adapts perturbation amplitude to system constraints—this parameter is not fixed but discovered per system.

4 Experiments

4.1 Synthetic Bench

We validate on a configurable synthetic bench (`bench-generic`) with planted ground-truth coupling. The bench hosts multiple virtual nodes in a single container, each exposed via HTTP. A topology file specifies:

- Node parameters (count, bounds, objective weights)
- Directed edges with coupling strength and channel
- Coupling shape (linear, threshold, saturation, polynomial)
- Noise model (Gaussian, colored, drift)

The bench computes objectives as $g(\mathbf{w}, \mathbf{n}) = \sum_k w_k \cdot h(n_k)$ where h is the shape function, then adds coupling from source nodes: $f_i = g_i + \sum_{j \to i} s_{ji} \cdot g_j^{\text{channel}}$. Noise is additive Gaussian per sample.

This provides exact ground truth: we know the planted coupling strength, shape, and topology, and compare against the detector's measured rising edge.

4.2 Experimental Design

Group A — Coupling strength sweep. Pair topology (2 nodes, 1 edge), linear shape, noise $\sigma = 0.02$. Coupling strength swept: $\{0.1, 0.2, 0.3, 0.5, 0.7, 0.9\}$. For linear coupling, the expected step amplitude at neutral hold ($n_k = 0.5$) pushed to extreme ($n_k = 1.0$) is $\Delta = s \times \sum_k w_k \times 0.5$.

Group B — Noise robustness. Pair topology, linear shape, coupling $s = 0.5$. Noise swept: $\sigma \in \{0.01, 0.02, 0.05, 0.10, 0.20\}$. Additional cells at block size 30 for noise = 0.10 to assess the observation budget tradeoff.

Group C — False positives. Pair topology, linear shape, coupling $s = 0.0$ (no edge). Noise: $\sigma \in \{0.02, 0.10, 0.20\}$. The protocol runs normally; the detector should find no coupling.

Group D — Nonlinear shapes. Pair topology, noise $\sigma = 0.02$. Shapes: saturation, threshold, polynomial. Each shape tested at coupling $\{0.7, 0.3\}$ to assess shape-dependence of the detection boundary.

4.3 Procedure

Each cell follows: dispatch the bench with planted topology, create two agents, let the coordinator cycle through push/pause/hold rounds until ≥ 200 trials, then query the CFAR detector endpoint for each ordered pair.

4.4 Prior Validation on Realistic Benches

Before the systematic sweep, the protocol was validated on two additional bench targets that model real-world complexity:

Greenhouse simulation. Two agents on coupled greenhouse controllers with 6+ cascaded non-linear transforms (thermal, CO2, humidity, crop-phase), crop model drift through growth phases, and per-sample SNR of approximately 0.67. At coupling 0.9, bidirectional coupling was detected across multiple objective channels. At coupling 0.0, zero false positives. All passive methods (FFT, Granger, mutual information, transfer entropy) failed on the same data at equivalent trial budgets. This is the hard case: deeply non-linear, non-stationary, noisy. Detection succeeds because active probing eliminates the dominant noise source (receiver self-exploration) and the block design creates a sustained step that survives the substrate's low-pass filtering.

Chain-4 composition. Four agents on a 4-node linear chain with edges at coupling 0.7, 0.5, and 0.3. All 3 edges detected with measured step amplitudes matching planted ground truth. Zero false positives across 12 evaluated pairs. This validates multi-agent turn-taking (4 agents sharing one lease sequentially) and confirms detection generalizes beyond pair topology.

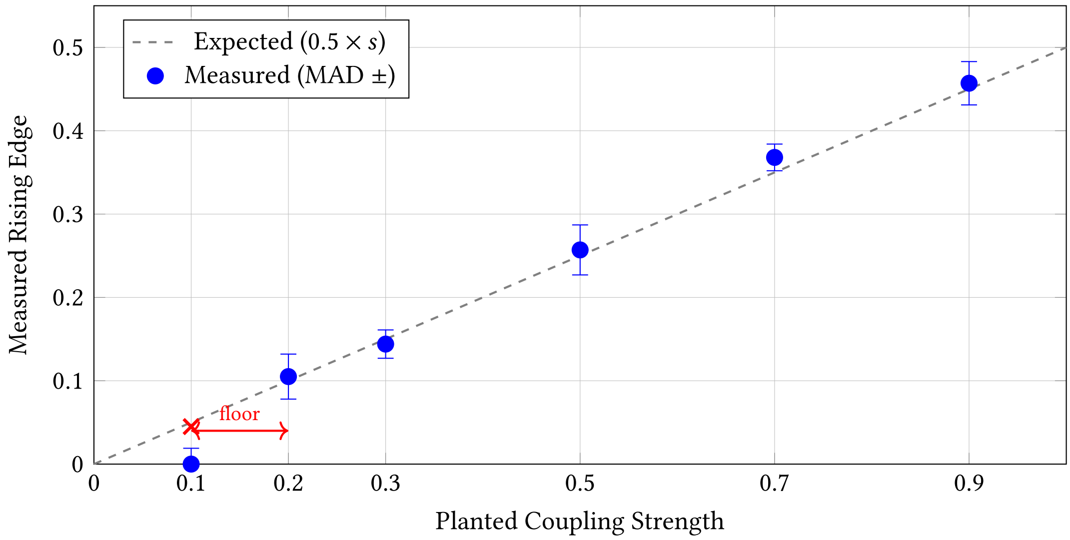

Figure 2: Detection sensitivity at noise $\sigma = 0.02$ (linear coupling). Blue circles: measured rising edge with MAD error bars. Dashed line: expected edge ($s \times 0.5$). Red ×: not detected (below CFAR threshold). Measurement error consistently below 5%. Detection floor lies between coupling 0.1 and 0.2.

Table 1: Linear coupling strength sweep at noise $\sigma = 0.02$.

Strength	Detected	Rising Edge	Expected	Error	MAD
0.1	**no**	0.045	0.05	—	0.019
0.2	yes	0.105	0.10	+5.2%	0.027
0.3	yes	0.144	0.15	−4.2%	0.017
0.5	yes	0.257	0.25	+2.9%	0.030
0.7	yes	0.368	0.35	+5.2%	0.016
0.9	yes	0.457	0.45	+1.4%	0.026

5 Results

5.1 Group A: Detection Sensitivity

The detection floor lies between coupling 0.1 and 0.2. At 0.1, the planted step amplitude (0.05) is comparable to the CFAR threshold ($k \times \text{MAD} \approx 3.0 \times 0.019 \approx 0.057$). The signal is present but below the noise-adaptive threshold. Above 0.2, detection is reliable with measurement error consistently below 5%.

5.2 Group B: Noise Robustness

Table 2: Noise robustness at coupling $s = 0.5$ (expected edge 0.25).

Noise σ	Block	Detected	Edge	MAD	Note
0.01	15	yes	0.237	0.034	
0.02	15	yes	0.257	0.030	(Group A cell)
0.05	15	yes	0.259	0.061	
0.10	15	no	—	0.124	signal < threshold
0.10	30	no	—	0.093	budget contrast: MAD drops, still below threshold
0.20	15	no	—	0.253	noise equals signal

Detection fails between noise 0.05 and 0.10. At noise 0.10, the asymptotic CFAR threshold ($k \to 3.0$ as $N \to \infty$) is 0.30, permanently above the signal (0.25). Doubling the block size from 15 to 30 reduces MAD by 25% and narrows the gap from 63% to 14%, but does not cross the threshold—the boundary is governed by the SNR, not the observation budget alone.

5.3 Group C: False Positives

Table 3: False positive test: coupling $s = 0.0$ (no edge exists).

Noise σ	Detected	MAD	Trials
0.02	no	0.017	2 rounds
0.10	no	0.116	1 round
0.20	no	0.136	6 rounds

Zero false positives across the full noise range, including extreme noise ($\sigma = 0.20$) where the MAD is an order of magnitude larger than at the standard operating point. The threshold is conservative.

5.4 Group D: Nonlinear Coupling Shapes

Table 4: Nonlinear shape generality at noise $\sigma = 0.02$.

Shape	Strength	Detected	Edge	MAD	Note
Saturation	0.7	yes	−0.113	0.028	inverted step
Threshold	0.7	yes	0.703	0.026	discontinuous jump
Polynomial	0.7	yes	0.517	0.028	quadratic step
Saturation	0.3	**no**	—	0.023	step compressed below floor
Threshold	0.3	yes	0.311	0.017	large step from discontinuity
Polynomial	0.3	**no**	—	0.005	step below floor

All three nonlinear shapes are detected at coupling 0.7. Saturation produces an inverted step (the response at extreme parameters is *lower* than at mid-range); the detector correctly identifies this via the absolute rising-edge test.

At coupling 0.3, shape matters: threshold coupling is detected (large step from the discontinuity), but saturation and polynomial coupling fall below the detection floor. This is expected—nonlinear shapes compress or redirect the step amplitude, and at low coupling the compressed signal is indistinguishable from noise.

6 Discussion

6.1 What the Protocol Contributes

The push/pause/hold protocol is the contribution—not the CFAR detector. CFAR is a well-established technique from radar signal processing [4]. What is new is the choreography: autonomous agents, each with their own objectives and unaware of each other, take turns as active perturbers and passive observers. The coordination is minimal—a scheduling constraint (one sender at a time), not information exchange. Agents do not share objectives, results, or detected edges.

Without the protocol, there is no signal for CFAR to detect. Two independently adjusting agents produce confounded objective trajectories that passive methods cannot separate [10]. The protocol converts the live system into a sequence of structured experiments.

6.2 Detection Boundaries Are Parameterized

The detection boundaries presented in Tables 1–2 are specific to the observation parameters listed in Section 3.5. The method has no fundamental sensitivity limit—the boundary is a function of observation budget (block sizes, number of rounds) and detector configuration (P_{fa}, minimum sample counts). The CFAR scale factor k decreases with more reference cells; the MAD estimate stabilizes with more samples.

We present results at one operating point ($B = 15$, $P_{\text{fa}} = 0.05$) with one budget contrast ($B = 30$ at noise 0.10). This characterizes where the method works and where it stops under reasonable observation costs. The noise failure at $\sigma = 0.10$ represents an SNR of approximately 2.5:1—an aggressive operating point for live systems.

6.3 Threshold Calibration

The CFAR threshold (2) is computed from per-sample noise (MAD of individual objective values), but the detection test compares block medians. The actual false alarm rate is therefore lower than the nominal $P_{\text{fa}} = 0.05$—the threshold is conservative. This is confirmed empirically: zero false positives were observed across all noise levels (Table 3). A median-aware threshold would improve sensitivity at high noise but requires re-deriving the false alarm rate for median statistics. We leave this to future work and report results with the conservative threshold.

6.4 Nonlinear Coupling

The protocol is coupling-shape-agnostic by design. The sender pushes to extreme parameters and returns to neutral; the detector asks whether the receiver's signal changed and recovered. The shape of the coupling function determines the step amplitude but not whether a step exists. All four tested shapes (linear, threshold, saturation, polynomial) were detected at adequate coupling strength (Table 4).

At low coupling (0.3), nonlinear shapes that compress the step (saturation, polynomial) fall below the detection floor, while threshold—which produces a large discontinuous jump—is detected. This is not a limitation of the protocol but a property of the specific coupling function's step response at the operating point.

6.5 Topology Limitation

The scanner discovers step-change coupling between node pairs. It does not distinguish direct edges from transitive paths. In a branching topology (e.g., $A \rightarrow B \rightarrow D$ and $A \rightarrow C \rightarrow D$), node D responds to A's perturbation through transitive paths, and the scanner would report a step change for the pair (A, D) even though no direct edge exists. Distinguishing direct from transitive coupling requires either known edge weights to subtract expected contributions or graph-theoretic analysis on the discovered step-change graph. This is an open problem for future work.

6.6 Positioning: Empirical Coupling Discovery

The method occupies a space parallel to existing causal frameworks. Pearl's causal hierarchy [8] formalizes inference from interventions designed by a central researcher who controls the experimental apparatus. System identification [7] estimates parameters within a model structure that is specified a priori.

Our approach runs parallel to Pearl's ladder rather than directly on it. Pearl's experimenter is centralized and human. Ours is distributed and autonomous. Pearl discovers Boolean causal arrows. We discover weighted, continuous, reversible step-changes. Pearl requires an external agent who intervenes. We have the agents themselves serving as both perturbers and sensors, coordinated through a shared lease. The coupling graph emerges from their through-operation—as a byproduct of tending a live system.

Pearl's framework is the special case where the experimenter is centralized and the model structure is hypothesized. System identification is the special case where the structure is known and only parameters are estimated. We are the case where the experimenter is distributed, autonomous, and the structure is discovered empirically, without specification.

7 Position and Future Work

7.1 What This Enables

Coupling detection is the foundation step of a larger program: building an empirically-discovered, maintainable coupling model of a live system. The push/pause/hold protocol produces a coupling scan—a directed graph of measured step responses between subsystems, valid for the moment it was taken. This scan is not an end in itself but the basis for:

Characterization. The probe data collected during detection already contains more information than the binary "coupling detected." Each edge carries a measured step amplitude, a baseline noise level, and a reversibility fraction. Extracting propagation delay, settling time, and response curve shape from the same data requires only signal processing—no additional probing. The scan becomes a characterized connectome: weighted, directional, per-channel.

Prediction. If measured edges compose along graph paths, the coupling graph becomes a forward model: "if node A changes by Δ, what is the predicted response at distant node D through the chain $A \rightarrow B \rightarrow C \rightarrow D$?" Whether edges compose linearly or require a learned composition function is the open question. Chain4 validation (4-node linear topology, 3 edges) showed measured edges match planted ground truth—but predicting a distant node's response from composed edges is not yet tested.

Maintenance. A frozen coupling scan serves as a reference model. The system drifts; the scan ages. Prediction error—comparing the model's expected response against actual behavior—localizes where structure changed. This triggers targeted rescanning of the affected neighborhood rather than a full system rescan. The ratio of scanning to operating adapts to structural volatility: stable systems need infrequent rescans, volatile systems need frequent ones.

Cooperation. Once agents understand their coupling, they can act on it. Two agents connected by a coupling edge currently find individually optimal but jointly suboptimal configurations. The coupling graph reveals tradeoffs invisible to independent agents. This requires no change to the detection protocol—only a consumer of the graph it produces.

7.2 Limitations

Topology. The scanner discovers step-change coupling between node pairs. It does not distinguish direct edges from transitive paths. In a branching topology ($A \rightarrow B \rightarrow D$ and $A \rightarrow C \rightarrow D$), node D responds to A's perturbation through transitive paths. The scanner reports a step change for (A, D) even though no direct edge exists. Distinguishing direct from transitive coupling is an open problem.

Threshold calibration. The CFAR threshold uses per-sample noise (MAD of individual values) but the detection test compares block medians. The actual false alarm rate is lower than the nominal P_{fa}—the threshold is conservative, confirmed empirically by zero false positives across all noise levels. A median-aware threshold would improve sensitivity but requires re-deriving the false alarm statistics.

Coordination scale. The lease-based turn-taking supports a small number of agents (tested with 2–4). Scaling to many agents requires group-based coordination (scoping lease tables by experiment tag) to avoid global contention. The current global coordination table is a prototype limitation, not a fundamental constraint.

Auto-calibration. Detection parameters—block sizes, push amplitude, cadence, P_{fa}—are currently fixed. Metaheuristic search could calibrate these per system, adapting the observation protocol to each substrate's characteristics. This is future work.

8 Conclusion

We presented a push/pause/hold coordination protocol that enables autonomous agents—unaware of each other, communicating only through their shared substrate—to discover coupling structure. The protocol is coupling-shape-agnostic and produces a reversible step signal that a CFAR detector evaluates against a noise-adaptive threshold.

Empirical validation on a synthetic bench with planted ground truth shows detection fails at coupling 0.1 and succeeds from 0.2 upward at noise $\sigma = 0.02$, with step amplitude measurements within 5% of ground truth. Zero false positives were observed across all noise levels tested. All tested nonlinear shapes (saturation, threshold, polynomial) were detected at adequate coupling.

The coupling scan produced by this protocol is the foundation for a larger program: characterized connectomes, forward prediction from composed edges, drift detection through prediction error, and cooperative agent behavior informed by discovered structure.

Reproducibility

All 21 experiment cells (raw detection JSON, observation parameters, and summary table) and the synthetic bench used to generate them are available at `https://github.com/godon-dev/godon` under `papers/detection/`. The coordination protocol (`godon-breeders`), CFAR detector (`godon-causal`), and bench target (`godon-bench-generic`) are in separate repositories within the same organization. All code is open-source under AGPLv3. Experiments are reproducible via GitHub Actions workflows that deploy the bench, create agents, run detection rounds, and query the detector endpoint.

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